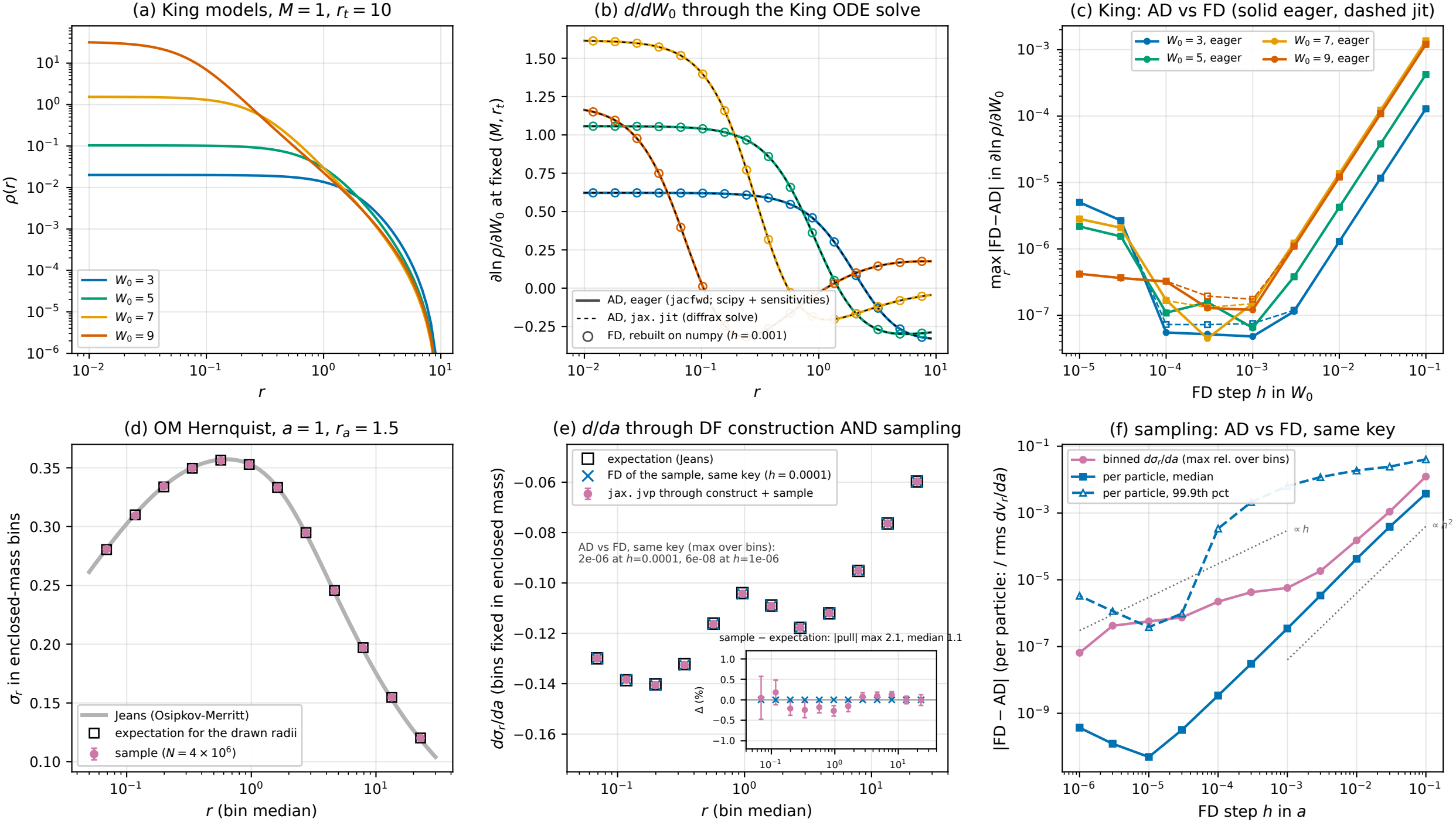

(a) King models, $M = 1$, $r_t = 10$
$\rho(r)$
r
$W_0 = 3$
$W_0 = 5$
$W_0 = 7$
$W_0 = 9$
(b) d/dW_0 through the King ODE solve
$\partial \ln \rho / \partial W_0$ at fixed (M, r_t)
r
AD, eager (jacfwd; scipy + sensitivities)
AD, jax.jit (diffrax solve)
FD, rebuilt on numpy ($h = 0.001$)
(c) King: AD vs FD (solid eager, dashed jit)
$\max_r |\mathrm{FD} - \mathrm{AD}|$ in $\partial \ln \rho / \partial W_0$
FD step h in W_0
$W_0 = 3$, eager
$W_0 = 5$, eager
$W_0 = 7$, eager
$W_0 = 9$, eager
(d) OM Hernquist, $a = 1$, $r_a = 1.5$
σ_r in enclosed-mass bins
r (bin median)
Jeans (Osipkov-Merritt)
expectation for the drawn radii
sample ($N = 4 \times 10^6$)
(e) d/da through DF construction AND sampling
$d\sigma_r/da$ (bins fixed in enclosed mass)
r (bin median)
expectation (Jeans)
FD of the sample, same key ($h = 0.0001$)
jax.jvp through construct + sample
AD vs FD, same key (max over bins):
2e-06 at h=0.0001, 6e-08 at h=1e-06
sample − expectation: |pull| max 2.1, median 1.1
Δ (%)
(f) sampling: AD vs FD, same key
$|\mathrm{FD} - \mathrm{AD}|$ (per particle: / rms dv_r/da)
FD step h in a
binned $d\sigma_r/da$ (max rel. over bins)
per particle, median
per particle, 99.9th pct
$\propto h$
$\propto h^2$